15.1 Exploring Sample Space

Lesson Introduction

Benchmark	MA.7.DP.2.1 Determine the sample space for a simple experiment.	Learning Targets	• I can determine all possible outcomes of a given scenario. • I can determine whether or not a simple experiment is fair.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.8.DP.2.1 Determine the sample space for a repeated experiment.

Essential Questions	• How does the number of possible outcomes for an experiment relate to the likelihood of a particular event? • How can we apply probability to inform a prediction?
Lesson Narrative	This lesson is the first within the unit, which introduces probability. The lesson engages students in thinking about sample space by using contexts they are already familiar with – games and fairness. While key vocabulary related to MA.7.DP.2.1 is introduced, this lesson is also designed to hook students by using their prior knowledge and experience relating to probability. In Lesson 2, students will delve more explicitly into determining sample spaces for simple experiments.

Component Summary	15.1.1 Warm-Up (~5 min)	Estimation activity using benchmark percentage of 50% (SE pg. 401)	15.1.3 Exploration (5 min)	Students discover how “fairness” relates to probability (SE pg. 404)
	15.1.2 Guided Instruction (10 min)	Introduction of key probability terms (SE pg. 402)	15.1.4 Activity (20-25 min)	Students compare sample spaces and fairness of carnival games in a gallery walk activity (SE pg. 404)
	15.1.5 Wrap-Up (~5 min)	Students reflect on their understanding of the lesson (SE pg. 405)		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Outcome • Sample space • Event <u>Additional Terms:</u> N/A	• Gallery Walk images • (Optional) Example game pieces to model gallery walk games	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may believe that all outcomes are equally likely in all scenarios.
Students may believe that future outcomes in these experiments depend on previous outcomes (all examples in this lesson are of independent probability).
Students may confuse the terms “outcome” and “event.”</td><td>In 15.1.3 Exploration, consider asking students to make the explicit connection between fractional sizes of the spinner sections and the “fairness” of the spinners.
Model choosing a card from Doug’s chore list to demonstrate the difference between “outcome” and “event.” Using repetition can help students solidify the vocabulary.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Number Talk
The 15.1.1 Warm-Up contains a question that asks students to estimate a percentage given a fraction and explain their reasoning. Allow each student time to independently come up with a strategy for estimating over or under 50%. Then, select students to share out different strategies they used in solving the problem. Consider adjusting the time for the Warm-Up to 10 minutes if utilizing this strategy.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
The 15.1.4 Activity is a collaboration opportunity for students. Students get to work together to review carnival game rules and determine whether or not those games should be considered “fair.” Student pairs also have the opportunity to discuss and reflect on their thinking with other student groups at the conclusion of the activity.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The carnival games presented in 15.1.4 require students to read the rules of a game and determine the possible outcomes and “fairness” of that game. Auditory learners would benefit from hearing the verbal description of the activities and both kinesthetic and visual learners would benefit from watching a demonstration of the game.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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15.1.1 Warm-Up: Carolyn Beatrice Parker

Component Overview: *Students practice using mental math strategies to assess a percentage value relative to the benchmark percentage of 50%.*

15.1.1 Warm-Up: Carolyn Beatrice Parker

Carolyn Beatrice Parker, born in Gainesville, FL in 1917, was the first African American woman known to receive a graduate degree in physics and subsequently work on the top-secret government project known as the Manhattan Project. Before earning her master's degree in physics, she earned a master's degree in mathematics and was also a teacher in Florida.

In 2020, an elementary school in Gainesville, FL was re-named in her honor. At this time, 11 of the 21 elementary schools in the Alachua County Public School system were named after people.

1. Without performing any calculations, select the statement that best describes the percentage of the elementary schools in Alachua County that were named after people as of 2020.

   - significantly less than 50%
   - slightly less than 50%
   - exactly 50%
   - slightly more than 50%
   - significantly more than 50%

2. Explain your reasoning.
   Answers will vary. The statement that best describes the percentage is slightly more than 50% because 21 is close to 20 and 10 is 50% of 20, therefore 11 is slightly more than 50% of 21.

Number Talk
Consider also allowing students to discuss their reasoning with a partner first before bringing the class back for a whole-group discussion. This practice can help students clarify their mathematical thinking.

- *Do you know any schools in your area named after someone famous?*

15.1.2 Guided Instruction: Determining Outcomes

Component Overview: *Students will begin practicing creating sample spaces to display the total possible outcomes for an experiment. They will identify the number of outcomes in a particular event. Activities are designed to build familiarity with probability vocabulary.*

15.1.2 Guided Instruction: Determining Outcomes

An ***outcome*** is a possible result of an experiment.

1. A spinner divided into four equal areas is shown.

   a. How many outcomes are there from spinning the spinner?
   4

In a probability model for a random process, the ***sample space*** is a list of the individual outcomes that are being considered.

   b. With your partner, discuss the sample space for the spinner. Write a list of the possible outcomes.
   B, R, G, Y

An ***event*** is a set of possible outcomes resulting from an experiment. In general, an event is any subset of a sample space.

   c. How many outcomes are included in the event that the spinner lands on blue?
   One

   d. Primary colors include yellow, blue, and red. How many outcomes are included in the event that the spinner lands on a primary color?
   Three

- *How do you find the total number of outcomes?*
- *Does one outcome seem more likely than another?*

Students are becoming familiar with probability vocabulary for the first time. To have additional practice understanding the concept of sample space, consider prompting students to discuss the sample spaces for different experiments: outcomes of a Miami Heat game, shoes to wear to school, rolling a die, flipping a coin, etc.

15.1.2 Guided Instruction: Determining Outcomes (continued)

2. Doug's parents created a set of cards with chores on them. Each Saturday, Doug chooses a card to determine his chore.

Mow the yard	Dust and vacuum the house	Clean the bathrooms	Clean your bedroom
Clean your bedroom	Mow the yard	Weed the garden	Mow the yard
Clean the bathrooms	Parents' choice	Dust and vacuum the house	Free day

   a. What are the possible outcomes for the chore cards?
      Mow the yard, Dust and vacuum the house, Clean the bathrooms, Clean your bedroom, Weed the garden, Parents' choice, Free day

   b. Which outcomes have more than one card?
      Mow the yard, Dust and vacuum the house, Clean the bathrooms, Clean your bedroom

   c. On a particularly hot Saturday, Doug is hoping he doesn't select an outdoor chore. Which outcomes could be included in the event that he chooses an outdoor chore?
      Mow the yard, Weed the garden

Students should be able to complete question 2 independently. Consider having them check in with a partner to compare responses.

- *Which outcome has the most cards?*
- *Which outcome(s) have the least cards?*

For English Language Learners, determining what constitutes an "outdoor" vs. "indoor" chore may be challenging to do independently. Consider providing a visual for each card as support for this activity.

Consider asking the question, "if Doug chooses a card at random, which chore do you think he is most likely to get?" Students can begin to make connections between the number of outcomes and the probability of an event.

15.1.2 Guided Instruction: Determining Outcomes (continued)

d. How many outcomes could be included in the event that Doug chooses an indoor chore?
6 - Dust and vacuum the house (2 cards), Clean the Bathrooms (2 cards), and Clean your bedroom (2 cards)

3. Use the word bank to complete the summary of key concepts below.

event	outcomes	sample space	set

To determine the ___*sample space*___ of a simple experiment, find all the ___*outcomes*___ that are possible and write them as a ___*set*___.

An ___*event*___ is a subset of the sample space, such as rolling an even number on a 6-sided die.

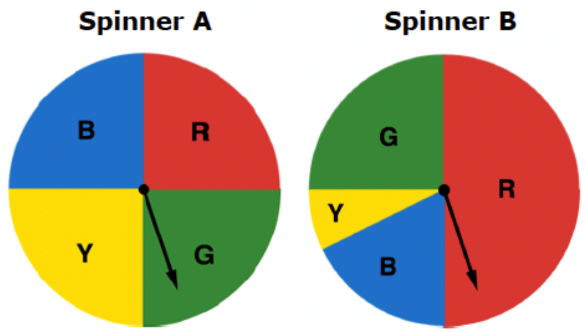
Clarify in question 3 that "set" and "list" can be used interchangeably in this context.

15.1.3 Exploration: Digging Deeper Into Outcomes

Component Overview: *Students will begin exploring the concept of "fairness" by comparing two spinners with the same total number of outcomes, but different likelihoods associated with each outcome.*

15.1.3 Exploration: Digging Deeper into Outcomes

Two spinners are shown.



1. Work with your partner to compare and contrast the features of the two spinners.

Similarities	Differences
size of the spinner *amount of sections* *name and colors of the sections*	*size of the sections except for G*

2. One of the spinners is considered a fair spinner. Complete the sentence by selecting the fair spinner and writing an explanation.

- Spinner A
- Spinner B

is the fair spinner because *all of the sections have the same chance to be selected*.

Allow students to discuss which spinner is more favorable for each outcome. For example, pose the question, "If you wanted a spinner to land on red, which spinner would you prefer to use?" Encourage students to use mathematical support for their responses.

- *What does it mean for the spinner to be "fair?"*
- *Can you give an example of a fair game?*
- *An example of a game that is not fair?*

15.1.4 Activity: Gallery Walk

Component Overview: *Students will analyze four different carnival games to determine the possible outcomes of each game and whether or not the game would be considered "fair." Post the activities around the room and allow students to circulate among them. Alternatively, partner pairs can receive copies of the carnival games to complete the activity.*

15.1.4 Activity: Gallery Walk

1. Work with your partner to determine the sample space for each game and whether the game is fair. Fill in the chart with your answers.

Game Name	Sample Space	Is it fair?
Skee-Ball	*0, 10, 20, 30, 40, 50, 100*	*No*
Card Draw	*Clubs- Ace,2,3,4,5,6,7,8,9,10, Jack, Queen, King* *Spades- Ace,2,3,4,5,6,7,8,9, 10, Jack, Queen, King* *Hearts- Ace,2,3,4,5,6,7,8,9, 10, Jack, Queen, King* *Diamonds- Ace,2,3,4,5,6,7,8, 9, 10, Jack, Queen, King*	*Yes*
Prize Wheel	*orange, blue, red, yellow, white*	*No*
Rubber Ducks	*1,2,3,4,5,6,7,8,9,10,11,12,13, 14,15,16,17,18,19,20,21,22, 23,24,25*	*Yes*

2. Which game has the largest sample space?
   Card Draw

3. Discuss with your partner which game's sample space was the most difficult to determine.
   Answers will vary.

This activity requires students to read the rules for a game to determine whether or not the game is fair. For students who struggle with reading comprehension, it may be helpful to provide a demonstration of how each game is played. Consider splitting the class into four groups to simulate a few rounds of each game.

Consider asking students to make rule proposals for the games they deemed unfair. How can the rules be modified to make these games fair?

Why do you think a carnival would choose to offer both fair and unfair games?

15.1.4 Activity: Gallery Walk (continued)

4. Carnivals include many games similar to these. What do you notice about the fairness of such games?
   Answers will vary. I notice that some games are noticeably fair or unfair, while other games are unnoticeably unfair.

5. Find another pair of partners to form a group of four. Compare your responses on whether each game is fair and discuss your reasoning. Record one similarity and one difference you notice in your reasoning.
   Answers will vary.

Similarity	Difference
We all thought that the Card Draw was fair because all of the suits had an equal chance of being drawn.	*We thought that the Prize Wheel was not fair because all of the colors are not the same size, and our partners thought that the Prize Wheel was fair because all of the colors are the same size.*

6. Future lessons in this unit will explore probabilities with fair coins, fair dice, and fair spinners. Describe what you think it means for a coin, a die, or a spinner to be fair.
   Answers will vary. Fair means that all of the outcomes have the same chance to be selected.

Provide an opportunity for meaningful reflection with questions 3 and 4 on "fairness." This response could be used for you to evaluate student understanding from the lesson.

15.1.5 Wrap-Up: Reflection

Component Overview: *Students complete a self-reflection on their level of understanding of the content of this lesson.*

15.1.5 Wrap Up: Reflection

Complete each statement based on your understanding at this point.
Answers will vary.

1. I am still unsure about ____________________________
   ____
   ____.

2. One question I have that will help me better understand
   is ____
   ____
   ____?

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.